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Editorial Office
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Dear Editor,

We are pleased to submit our manuscript entitled **"Quantum Agrivoltaic Digital Twin: Coordinated Vibronic Light-Harvesting, Environmental Calibration, and Agro-Rural Cybersecurity"** for consideration as a Research Article in *Nature Energy*.

Summary of the work

Agrivoltaics — co-locating photovoltaics and crops — typically relies on passive light reduction, which limits agricultural productivity. We present a "quantum-to-organism" framework where an organic photovoltaic (OPV) layer integrated with a nanoparticle-on-mirror (NPoM) plasmonic cavity actively shapes the light spectrum to protect quantum coherence in the crop's light-harvesting complex, while simultaneously enabling label-free, single-molecule-sensitivity diagnostics.

Our key findings include:

- **Coherence-Protected Dynamics and Diagnostics:** Dual-band OPV filtering extends FMO excitonic coherence lifetimes by 50 % and transport yields by 25 % at room temperature. We resolve the NPoM plasmonic transport trade-off, identifying an optimal mode volume of $1.2\,\mathrm{nm}^3$ that preserves crop productivity while enabling SERS readout.
- **Greenhouse Microclimate and Economics:** A $500\,\mathrm{m}^2$ Cameroonian cooperative greenhouse model demonstrates a 28 % water saving ($650\,\mathrm{m}^3\,\mathrm{yr}^{-1}$), $90\,\mathrm{MWh}\,\mathrm{yr}^{-1}$ of electricity generation, and a 4.23 yr payback period under a 30 % blended-finance grant.
- **Agro-Rural Cybersecurity:** Field telemetry is secured via room-temperature decoy-state BB84 QKD, with low-latency (3.7 ms) edge inference deployed on ARM Cortex-M4 IoT nodes.

Why *Nature Energy*?

Nature Energy is the ideal venue for this work, which occupies the intersection of quantum materials, photovoltaics, microclimate physics, and socioeconomic policy. Our "quantum-to-organism" framework bridges angstrom-scale vibronic couplings with field-level agricultural yields, showing that active quantum coherence protection can provide quantifiable sustainability and economic benefits. The integration of spectral coherence protection, SERS diagnostics, and QKD-secured IoT within a unified agrivoltaic platform is, to our knowledge, not previously reported, and aligns directly with the journal's mission.

Prior discussions and related submissions

A companion manuscript focusing on selective vibronic excitation in isolated FMO complexes has been submitted to the *Journal of Physical Chemistry Letters* (Manuscript ID jz-2026-00994t, revision submitted June 20, 2026). The present manuscript extends that physical framework to a complete agrivoltaic system, introducing the NPoM cavity, SERS readout, dynamic Stark detuning, farm-scale microclimates, lifecycle assessments, cooperative economics, and QKD cybersecurity. No part of this manuscript is under consideration elsewhere.

All authors have approved the manuscript and agree with its submission. The authors declare no competing financial or non-financial interests.

Suggested reviewers:

- Prof. Alexandra Olaya-Castro (UCL, `a.olaya@ucl.ac.uk`)
- Prof. Joel Yuen-Zhou (UCSD, `joelyuen@ucsd.edu`)
- Prof. Jeremy J. Baumberg (University of Cambridge, `jjb12@cam.ac.uk`)
- Prof. Greg A. Barron-Gafford (University of Arizona, `gregbg@email.arizona.edu`)

Excluded reviewers: None.

Sincerely,

Steve Cabrel Teguia Kouam
(On behalf of all authors)

Enclosures: Manuscript, Supporting Information, Figures, and Graphical Abstract.